

13.3 Tukey's studentized range

For the family of all pairwise differences $\mu_k - \mu_l$, Tukey saw that the largest standardized error is the range of the estimated means divided by an estimate of their common standard deviation. When the means are estimated independently and with equal precision, as in a balanced one-way layout, this ratio is pivotal, and its upper α point gives intervals for all pairs with simultaneous coverage exactly $1 - \alpha$, shorter than Scheffé's or Bonferroni's.

13.3.1 The studentized range distribution

Definition 13.3.1. Studentized range

Let $Z_1, \dots, Z_g$ be independent $N(0, 1)$ variables, $g \geq 2$, and let $V \sim \chi^2(\nu)$ be independent of them. The law of

$$Q = \frac{\max_k Z_k - \min_k Z_k}{\sqrt{V/\nu}}$$

is the **studentized range distribution** with parameters g and ν . Its upper α point is written $q_\alpha(g, \nu)$.

The numerator is the range of a normal sample of size g . Dividing by $\sqrt{V/\nu}$ replaces the known standard deviation 1 by an independent estimate of it with ν degrees of freedom, just as a t variable is a studentized normal. For $g = 2$ the range is $|Z_1 - Z_2|$, a $N(0, 2)$ variable in absolute value, so $Q/\sqrt{2}$ is the absolute value of a $t(\nu)$ variable and

$$q_\alpha(2, \nu) = \sqrt{2} t_{\nu, \alpha/2}. \quad (13.3.1)$$

For larger g the distribution function is a double integral: conditioning on the smallest Z_k gives $\Pr(\max_k Z_k - \min_k Z_k \leq w) = g \int \varphi(z)[\Phi(z+w) - \Phi(z)]^{g-1} dz$, and one further integration over V gives the distribution of Q . SciPy evaluates it as `scipy.stats.studentized_range`. For $g = 5$ and $\nu = 20$ it gives $q_{0.05}(5, 20) = 4.232$, and 200000 simulated values of Q put the upper 5% point at 4.230.

```
import itertools
import numpy as np
from scipy import stats

alpha, reps = 0.05, 40_000

g, nu = 5, 20
rng = np.random.default_rng(20260919)
Z = rng.normal(size=(200_000, g))
V = rng.chisquare(nu, size=200_000)
Q = (Z.max(axis=1) - Z.min(axis=1)) / np.sqrt(V / nu)    # studentized range
q_sim = np.quantile(Q, 1 - alpha)
q_exact = stats.studentized_range.ppf(1 - alpha, g, nu)
print(f"upper 5% point of the studentized range, g = {g}, nu = {nu}:"
      f" simulated {q_sim:.3f}, scipy {q_exact:.3f}")
```

[Run this code online](#)

13.3.2 Tukey's intervals for a balanced layout

Consider the one-way model with g levels and the same number m of observations at every level, so $n = gm$, and assume normal errors. The level means $\bar{y}_1, \dots, \bar{y}_g$ are the least squares estimates of $\mu_1, \dots, \mu_g$, the pooled variance is $s^2 = \text{SSE}/\nu$ with $\nu = g(m-1)$, and the standard error of every difference $\bar{y}_k - \bar{y}_l$ is $s\sqrt{2/m}$.

Lemma 13.3.1. A contrast is bounded by the range

If $\sum_k c_k = 0$, then for all real $x_1, \dots, x_g$,

$$\left| \sum_k c_k x_k \right| \leq \frac{1}{2} \sum_k |c_k| (\max_k x_k - \min_k x_k),$$

with equality when $\mathbf{c}$ is a multiple of a pairwise difference $\mathbf{e}_k - \mathbf{e}_l$ and x_k, x_l are the largest and smallest of the x 's.

Proof. Let $a = \frac{1}{2}(\max_k x_k + \min_k x_k)$. Because $\sum_k c_k = 0$, $\sum_k c_k x_k = \sum_k c_k (x_k - a)$, and $|x_k - a| \leq \frac{1}{2}(\max_k x_k - \min_k x_k)$ for every k . The triangle inequality gives the bound. For $\mathbf{c} = \mathbf{e}_k - \mathbf{e}_l$, the left side is $|x_k - x_l|$ and the right side is the range, which agree when x_k, x_l are the extremes. ■

Theorem 13.3.2. Tukey's simultaneous intervals

In the balanced one-way model with normal errors, $g \geq 2$ and $\nu = g(m-1) \geq 1$, let $c_T = q_\alpha(g, \nu)$. Then:

(a) with probability exactly $1 - \alpha$,

$$\mu_k - \mu_l \in \left[\bar{y}_k - \bar{y}_l - c_T \frac{s}{\sqrt{m}}, \bar{y}_k - \bar{y}_l + c_T \frac{s}{\sqrt{m}} \right] \quad \text{for all } k \neq l; \quad (13.3.2)$$

(b) on the same event, for every contrast $\psi = \sum_k c_k \mu_k$,

$$\left| \sum_k c_k \bar{y}_k - \psi \right| \leq c_T \frac{s}{\sqrt{m}} \cdot \frac{1}{2} \sum_k |c_k|,$$

so these intervals for all contrasts also have simultaneous coverage exactly $1 - \alpha$.

Proof. By Theorem 7.5.1, the means $\bar{y}_k$ are independent $N(\mu_k, \sigma^2/m)$, and they are independent of s^2 , with $\nu s^2/\sigma^2 \sim \chi^2(\nu)$. Put $Z_k = \sqrt{m}(\bar{y}_k - \mu_k)/\sigma$ and $V = \nu s^2/\sigma^2$. These satisfy the assumptions of Definition 13.3.1, whatever the μ_k are. Now

$$\max_{k \neq l} |(\bar{y}_k - \bar{y}_l) - (\mu_k - \mu_l)| = \max_{k, l} [(\bar{y}_k - \mu_k) - (\bar{y}_l - \mu_l)] = \frac{\sigma}{\sqrt{m}} (\max_k Z_k - \min_k Z_k),$$

so all intervals Eq. 13.3.2 cover iff $(\max_k Z_k - \min_k Z_k)/\sqrt{V/\nu} \leq c_T$, an event of probability $1 - \alpha$, since the studentized range has a continuous distribution. For (b), apply Lemma 13.3.1 with $x_k = \bar{y}_k - \mu_k$: on that event the range of the x_k is at most $c_T s/\sqrt{m}$. The pairwise

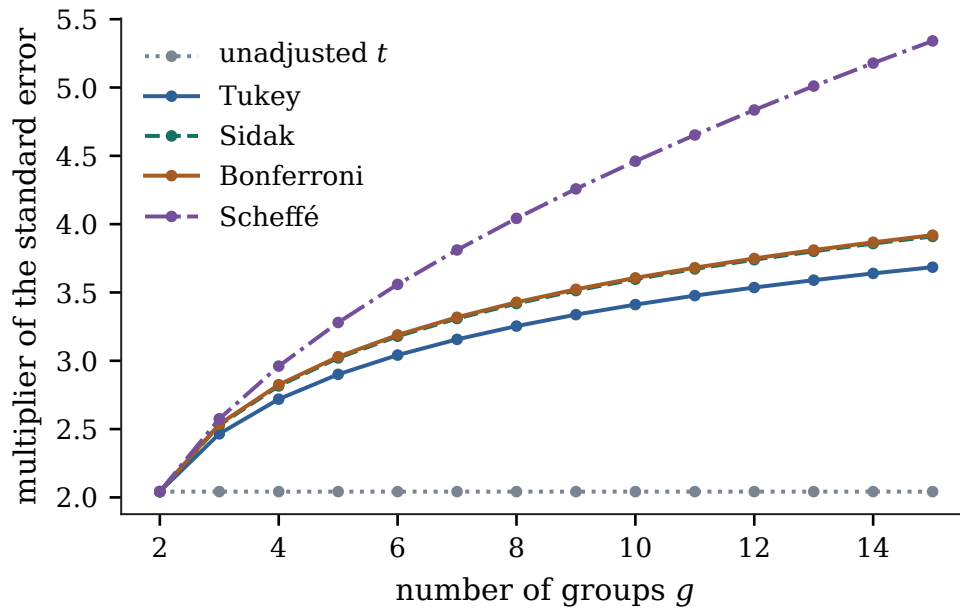


Figure 13.3.1: Multipliers of the standard error for 95% simultaneous intervals for all pairwise differences of g balanced means, $\nu = 30$ error degrees of freedom. Tukey's multiplier is $q_{0.05}(g, 30)/\sqrt{2}$. The Šidák and Bonferroni curves nearly coincide.

differences belong to the family of contrasts, so the coverage of the enlarged family cannot exceed that of the pairs, and it equals $1 - \alpha$.

Proposition 13.3.3. Tukey's intervals are the shortest for all pairs

In the balanced one-way model, suppose intervals $\bar{y}_k - \bar{y}_l \pm c s \sqrt{2/m}$ with a common multiplier c have simultaneous coverage at least $1 - \alpha$ for all pairs. Then $c \geq q_\alpha(g, \nu)/\sqrt{2}$. In particular the Bonferroni, Šidák and Scheffé intervals for all pairs are at least as long as Tukey's.

Proof. By the proof of [Theorem 13.3.2](#), the simultaneous coverage of such intervals is $\Pr(Q \leq c\sqrt{2})$, where Q has the studentized range distribution. This is at least $1 - \alpha$ only if $c\sqrt{2} \geq q_\alpha(g, \nu)$. The other three methods give intervals of this form with coverage at least $1 - \alpha$ (Theorem 13.2.2 and [Section 13.4](#)).

[Figure 13.3.1](#) compares the multipliers of the standard error of a difference for $\nu = 30$. With $g = 5$ groups they are 2.901 for Tukey, 3.021 for Šidák, 3.030 for Bonferroni and 3.280 for Scheffé. With $g = 10$ they are 3.411, 3.597, 3.607 and 4.461. The unadjusted multiplier is 2.042 throughout. For $g = 2$ all methods coincide, by [Eq. 13.3.1](#). Scheffé's method falls further behind as g grows, because it covers a space of dimension $g - 1$ while the family of pairs is finite.

```
import numpy as np
from scipy import stats
```

```

alpha, nu = 0.05, 30

def pairwise_multipliers(g, nu, alpha=0.05):
    m = g * (g - 1) // 2                                # number of pairs
    return {
        "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
        "Tukey": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
        "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
        "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
        "Scheffe": np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu)),
    }

for g in (2, 3, 5, 10):
    row = pairwise_multipliers(g, nu)
    print(f"g = {g:2d}: " + " ".join(f"{k} {v:.3f}" for k, v in row.items()))

```

[Run this code online](#)

For contrasts other than pairs the comparison can reverse. Part (b) of [Theorem 13.3.2](#) gives the half-width $q_\alpha(g, \nu) (s/\sqrt{m}) \frac{1}{2} \sum_k |c_k|$, while Scheffé's is $\sqrt{(g-1)F_\alpha(g-1, \nu)} (s/\sqrt{m}) (\sum_k c_k^2)^{1/2}$. With $g = 6$ and $\nu = 30$, in units of $s/\sqrt{m}$, a pairwise difference has half-width 4.301 by Tukey's method and 5.033 by Scheffé's; for $c = \frac{1}{3}(1, 1, 1, -1, -1, -1)$ the half-widths are 4.301 and 2.906. Tukey's bound is sharp for pairs and crude for contrasts spread over many means.

13.3.3 Unbalanced layouts: the Tukey–Kramer intervals

With unequal group sizes n_k the maximum standardized difference is no longer a studentized range. Tukey and Kramer proposed keeping the multiplier and using the standard error of each difference:

$$\bar{y}_k - \bar{y}_l \pm \frac{q_\alpha(g, \nu)}{\sqrt{2}} s \sqrt{\frac{1}{n_k} + \frac{1}{n_l}}, \quad \nu = n - g. \quad (13.3.3)$$

Tukey conjectured, and Hayter (1984) proved, that the simultaneous coverage of these intervals is never less than $1 - \alpha$, whatever the group sizes. The proof is beyond this chapter. How conservative the intervals are is a separate question; the simulation below estimates the coverage of 95% intervals in three designs from 40000 data sets each (standard error about 0.0011):

design	coverage
balanced, five groups of 5	0.9511
the education example (sizes 13 to 248)	0.9562
four groups of 2 and four of 40	0.9614

The balanced design is exact; the unbalanced ones overcover only modestly, even with very different group sizes.

```

import itertools
import numpy as np
from scipy import stats

```

```

alpha, reps = 0.05, 40_000

def tukey_kramer_coverage(n_k, reps, alpha, seed):
    """Share of data sets in which every Tukey-Kramer interval covers its difference
    . """
    rng = np.random.default_rng(seed)
    n_k = np.asarray(n_k)
    g, nu = len(n_k), n_k.sum() - len(n_k)
    q = stats.studentized_range.ppf(1 - alpha, g, nu)
    # group means (true means 0) and an independent pooled variance, as in the model
    means = rng.normal(size=(reps, g)) / np.sqrt(n_k)
    s = np.sqrt(rng.chisquare(nu, size=reps) / nu)
    ok = np.ones(reps, dtype=bool)
    for k, l in itertools.combinations(range(g), 2):
        half = q / np.sqrt(2) * s * np.sqrt(1 / n_k[k] + 1 / n_k[l])
        ok &= np.abs(means[:, k] - means[:, l]) <= half
    return ok.mean()

balanced = tukey_kramer_coverage([5] * 5, reps, alpha, seed=1)
education = tukey_kramer_coverage([13, 52, 248, 187, 90, 227, 127], reps, alpha,
    seed=2)
lopsided = tukey_kramer_coverage([2, 2, 2, 2, 40, 40, 40, 40], reps, alpha, seed=3)
print(f"balanced (5 x 5):           {balanced:.4f}")
print(f"education design:          {education:.4f}")
print(f"four groups of 2, four of 40: {lopsided:.4f}")

```

[Run this code online](#)

Example. All pairs of education levels

In Example (Education and the perceived position of a candidate) there are $g = 7$ education levels, so 21 pairs, and $\nu = 937$. The Tukey-Kramer multiplier is $q_{0.05}(7, 937)/\sqrt{2} = 4.179/\sqrt{2} = 2.955$. It declares 9 of the 21 differences nonzero, against 14 for unadjusted t intervals (Section 13.6 compares all the methods). Figure 13.3.2 shows the intervals. The significant differences all separate a lower education level from a higher one, and the three most educated levels do not differ from one another. The smallest group, with only 13 respondents, has the highest mean, but its intervals are so wide that only its differences from levels 5, 6 and 7 are established.

```

import itertools
import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                           # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu           # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu)) # Scheffe multiplier,

```

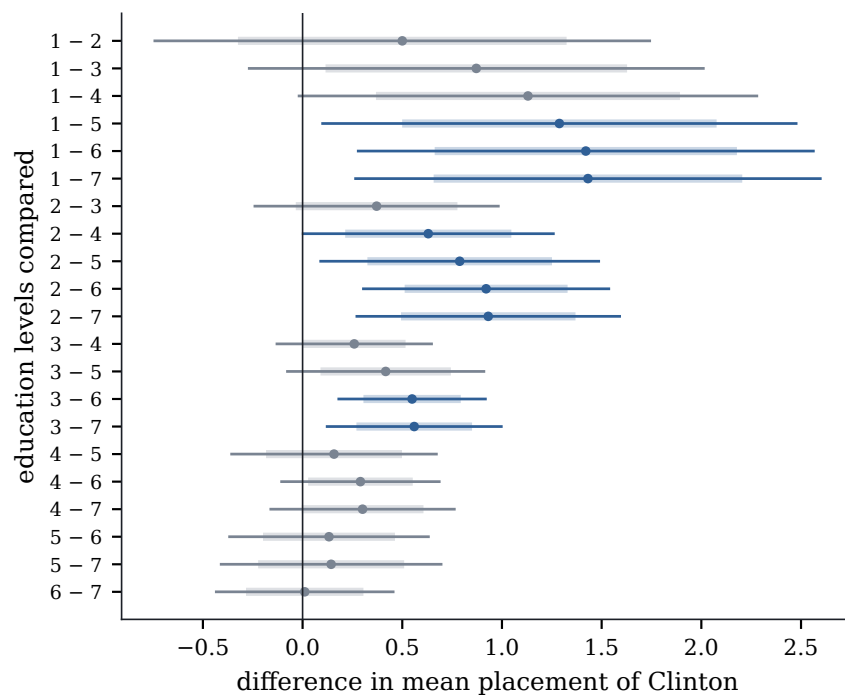


Figure 13.3.2: Differences in mean placement of Clinton between education levels, with 95% Tukey–Kramer simultaneous intervals (thin lines) and unadjusted t intervals (thick pale bands). Differences whose Tukey–Kramer interval excludes zero are drawn in blue.

```

q = g - 1

pairs = list(itertools.combinations(range(g), 2))
m = len(pairs)                                     # 21 comparisons
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
pval = 2 * stats.t.sf(np.abs(t), nu)

mult = {                                           # half-width = multiplier x se
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
    "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
    "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
    "Tukey-Kramer": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
    "Scheffe": c_sch,
}
for name, c in mult.items():
    print(f"{name:13s} multiplier {c:.3f}    pairs declared different: {np.sum(np.abs(t) > c)}")

```

[Run this code online](#)

13.3.4 Beyond the one-way layout

The proof of [Theorem 13.3.2](#) used only that the g estimates are independent, with a common variance $\sigma^2 c$ for a known c , and independent of s^2 . That also holds for the levels of one factor in a balanced multi-way design ([Exercise \(B1\)](#)), the subject of [Chapters 15](#) and [16](#). For correlated estimates, such as regression coefficients or adjusted means, neither form is exact; the exact critical value for any planned family then comes from a multivariate t distribution ([Exercise](#)

(C1)), which standard software computes.

13.3.5 Exercises

13.3.5.1 A. Check your understanding

Exercise. A1

Prove [Eq. 13.3.1](#), and deduce that Tukey's intervals for $g = 2$ are the ordinary t intervals.

13.3.5.2 B. Practice

Exercise. B1

In a two-way additive model $E(y_{ijk}) = \mu + \alpha_i + \beta_j$ with a rows, b columns and N observations in every cell, show that the column means $\bar{y}_{.j}$ are independent, each with variance $\sigma^2/(aN)$, and independent of the residual mean square. Deduce Tukey intervals for all differences $\beta_j - \beta_{j'}$, and state their degrees of freedom.

Solution. Each $\bar{y}_{.j}$ averages the aN observations of column j , and different columns use disjoint sets of independent observations, so the column means are independent with variance $\sigma^2/(aN)$. Their expectations are $\mu + \bar{\alpha} + \beta_j$, with $\bar{\alpha}$ the average row effect, so differences estimate $\beta_j - \beta_{j'}$. The column means are functions of the fitted values (the fitted value of cell (i, j) is $\bar{y}_{i..} + \bar{y}_{.j} - \bar{y}$, and averaging it over i gives $\bar{y}_{.j}$), so they are independent of s^2 by Theorem 7.5.1. The residual degrees of freedom are $\nu = abN - a - b + 1$, and the intervals are $\bar{y}_{.j} - \bar{y}_{.j'} \pm q_\alpha(b, \nu) s / \sqrt{aN}$, exact for all pairs of columns.

Exercise. B2

Treatments $1, \dots, g-1$ are to be compared with a control, level 0, in a balanced layout. Show that the exact simultaneous multiplier for the $g-1$ differences $\mu_k - \mu_0$ is the upper α point of $\max_{k \geq 1} |Z_k - Z_0| / \sqrt{V/\nu}$, and that it is smaller than $q_\alpha(g, \nu)$.

Solution. As in the proof of [Theorem 13.3.2](#), the largest standardized error among the differences with the control is $(\sigma/\sqrt{m}) \max_{k \geq 1} |Z_k - Z_0|$, and the statistic $D = \max_{k \geq 1} |Z_k - Z_0| / \sqrt{V/\nu}$ is pivotal. Let Q be the studentized range of $Z_0, \dots, Z_{g-1}$, so that $D \leq Q$. The event $\{D \leq q_\alpha(g, \nu) < Q\}$ has positive probability: it contains an open set of values of $(Z_0, \dots, Z_{g-1}, V)$, for instance those with Z_0 near the middle of the others and the range just above the critical value, and these variables have a positive joint density. Hence $\Pr(D > q_\alpha(g, \nu)) = \alpha - \Pr(D \leq q_\alpha(g, \nu) < Q) < \alpha$, and since D has a continuous distribution its upper α point is smaller than $q_\alpha(g, \nu)$. This is Dunnett's many-to-one procedure.

Exercise. B3

In the balanced one-way layout, the **studentized range test** rejects $\mu_1 = \dots = \mu_g$ when $Q_{\text{obs}} = (\max_k \bar{y}_k - \min_k \bar{y}_k) / (s/\sqrt{m}) > q_\alpha(g, \nu)$. Show that it has exact level α , and that

it rejects iff some Tukey interval Eq. 13.3.2 excludes 0. Show that the F test rejects iff $\sum_k (\bar{y}_k - \bar{y})^2 > (g-1)F_\alpha(g-1, \nu) s^2/m$, and describe, for $g = 3$, configurations of the means that one test rejects and the other does not.

Solution. Under the hypothesis Q_{obs} has the studentized range distribution (proof of Theorem 13.3.2 with all μ_k equal), so the level is exactly α , and $Q_{\text{obs}} > q_\alpha(g, \nu)$ says that the extreme pair's interval excludes 0; every other pair has a smaller difference. The F statistic is $m \sum_k (\bar{y}_k - \bar{y})^2 / \{(g-1)s^2\}$. For $g = 3$ and a fixed range $2a$, the sum of squares is $2a^2$ for means $(-a, 0, a)$ and $8a^2/3$ for $(-a, -a, a)$. In the limit $\nu \rightarrow \infty$, in units of $s/\sqrt{m}$, the range test rejects when $2a > 3.314$ and the F test when the sum of squares exceeds 5.991: so $(-a, 0, a)$ with $1.657 < a < 1.731$ is rejected only by the range test, and $(-a, -a, a)$ with $1.499 < a < 1.657$ only by the F test. The range test is the natural gate for pairs, the F test for contrasts in general.

13.3.5.3 C. Going deeper

Exercise. C1

Let $\lambda_1, \dots, \lambda_k \in \mathcal{C}(\mathbf{X}^\top)$ be chosen in advance, with t statistics $T_j = (\lambda_j^\top \hat{\beta} - \lambda_j^\top \beta) / \text{se}(\lambda_j^\top \hat{\beta})$. Show that $(T_1, \dots, T_k) = \mathbf{Z} / (s/\sigma)$, where $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{R})$ is independent of s , and $\mathbf{R}$ is the correlation matrix with entries $\lambda_i^\top \mathbf{G} \lambda_j / \sqrt{\lambda_i^\top \mathbf{G} \lambda_i \cdot \lambda_j^\top \mathbf{G} \lambda_j}$. Deduce that the constant c with $\Pr(\max_j |T_j| \leq c) = 1 - \alpha$ depends only on $\mathbf{R}$ and ν , and that it is at most the Bonferroni multiplier and at most Scheffé's multiplier for the span of the λ_j .